

Released Form

Published February 2026

Grade 5 Mathematics

North Carolina End-of-Grade Assessment



Public Schools of North Carolina

Department of Public Instruction | State Board of Education

Division of Accountability Services/North Carolina Testing Program



- 1 Lisa's puppy has a heart rate of 160 beats per minute. How many times will the puppy's heart beat in 60 minutes?
 - A 660
 - B 960
 - C 6,600
 - D 9,600

- 2 Brandon has $12\frac{2}{3}$ feet of wood. Steve has $7\frac{5}{6}$ feet of wood. How many feet of wood do they have altogether?
 - A $19\frac{1}{2}$
 - B $19\frac{3}{4}$
 - C $20\frac{1}{2}$
 - D $20\frac{3}{4}$

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3 Betty reads $\frac{1}{10}$ of her book every day. How much of her book will she have read after 4 days?

A $\frac{3}{10}$

B $\frac{2}{5}$

C $\frac{1}{2}$

D $\frac{3}{5}$

4 Which choice ***always*** describes a rhombus?

A a quadrilateral with equal side lengths and two pairs of parallel sides

B a quadrilateral with one pair of parallel opposite sides and one pair of nonparallel sides

C a quadrilateral with two pairs of unequal and parallel sides

D a quadrilateral with no parallel sides



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- 5 Ms. Owens bought a 5-pound bag of sand for a science class. She wants to give $\frac{1}{5}$ pound to each of her students. What is the maximum number of students that can receive $\frac{1}{5}$ pound of sand?

A 10
B 16
C 25
D 50

- 6 An expression is shown.

$$(5 \times 8 + 4) \div 2$$

What will happen to the value of the expression if both parentheses are removed?

A The value of the expression will remain the same.
B The value of the expression will decrease by 8.
C The value of the expression will increase by 12.
D The value of the expression will increase by 20.



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7 Mr. Barton has a piece of wood that is $3\frac{1}{4}$ feet long. He painted $\frac{1}{2}$ of the length black. What length of the wood did Mr. Barton paint black?

A $\frac{5}{8}$ foot

B $1\frac{5}{8}$ feet

C $1\frac{7}{8}$ feet

D $2\frac{3}{4}$ feet

8 What expression shows half the difference of 34 and 6?

A $6 - 34 \div \frac{1}{2}$

B $34 - 6 \div \frac{1}{2}$

C $(34 - 2) \div 6$

D $(34 - 6) \div 2$



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- 9 An art teacher has 1,032 pencils in boxes. Each box has 24 pencils. How many boxes of pencils does the art teacher have?
- A 42
 - B 43
 - C 44
 - D 45
- 10 Juan had $\frac{2}{3}$ pound of grapes. He gave $\frac{1}{6}$ pound of grapes to his friend. What fraction of a pound of grapes remains?
- A $\frac{1}{2}$
 - B $\frac{1}{3}$
 - C $\frac{1}{9}$
 - D $\frac{1}{18}$



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- 11 Rachel has wooden boards. They are each $2\frac{1}{4}$ feet long. What is the length of $3\frac{1}{2}$ wooden boards?

A $5\frac{3}{4}$ feet
 B $6\frac{3}{8}$ feet
 C $7\frac{7}{8}$ feet
 D 8 feet

- 12 What is the value of this expression?

$$\frac{1}{2} + \frac{2}{8} + \frac{3}{4}$$

A $\frac{6}{14}$
 B $\frac{6}{8}$
 C $1\frac{1}{2}$
 D $4\frac{1}{2}$



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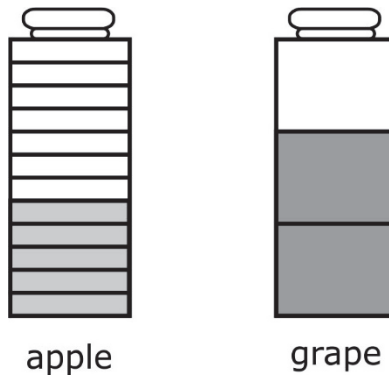
- 13 Cathy found a box that contained 720 marbles. She divided the marbles evenly among her 16 friends. How many marbles did each friend receive?
- A 45
 - B 46
 - C 47
 - D 48
- 14 A group of 6 friends picked a total of 10 pounds of strawberries. The strawberries were divided equally among the friends. What amount of strawberries did each friend get?
- A $\frac{3}{5}$ pound
 - B $1\frac{1}{16}$ pounds
 - C $1\frac{2}{5}$ pounds
 - D $1\frac{2}{3}$ pounds



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- 15 The shaded parts of these diagrams show how much juice was left after a party.



How much **larger** is the fraction of leftover grape juice than the fraction of leftover apple juice?

- A $\frac{1}{12}$
- B $\frac{1}{4}$
- C $\frac{4}{15}$
- D $\frac{4}{9}$

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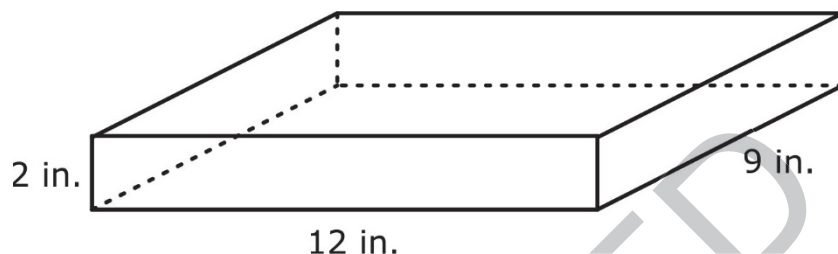


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Questions 16 through 20 require you to write your answers in the boxes. Write only one number or symbol in each box.

- 16 What is the volume of this figure, in cubic inches?



Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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- 17 There are 936 calories in 12 eggs. How many calories are in 1 egg?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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- 18 What is the value of n in this equation?

$$\frac{3}{8} + n = 1\frac{1}{4}$$

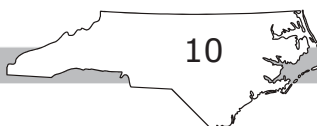
Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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- 19 The screen of a rectangular computer tablet is $\frac{1}{2}$ of a foot long and $\frac{3}{4}$ of a foot wide. What is the size of the screen, in square feet?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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- 20 An animal shelter uses 38 pounds of dog food per day. How many pounds of food will the shelter use in 365 days?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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This is the end of the calculator inactive test questions.

Directions:

- 1. Look back over your answers for the calculator inactive questions. You will not be able to go back and work on these questions once you are given a calculator.**
- 2. Make sure all your answers are entered appropriately in your book.**
- 3. Raise your hand to let your teacher know you are ready to begin the calculator active test questions.**
- 4. Do not begin work on the calculator active test questions until your teacher has given you a calculator.**

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Questions 21 through 25 require you to write your answers in the boxes. Write only one number or symbol in each box.

- 21 Adriana has 4 cups of peanuts. How many $\frac{1}{4}$ -cup servings does she have?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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- 22 A library receives 4,568 returned books in 4 weeks. It receives the same number of books each week. How many books does the library receive each week?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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- 23 A rectangle has a length of 10.42 meters and a width of 7.67 meters. What is the perimeter of the rectangle (in meters)?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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- 24 Teresa bought $\frac{3}{4}$ of a pound of chocolate fudge. She gave $\frac{1}{3}$ of that fudge to some friends. What fraction of a pound of the chocolate fudge did Teresa give to her friends?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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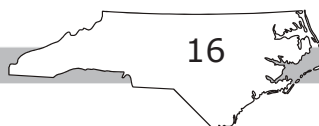


- 25 A group of 7 students share $\frac{1}{2}$ of a box of erasers. How much of the box does each student receive?

Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and / are allowed in your answer. Mixed numbers are entered by adding a space after the whole number. Spaces are only allowed between whole numbers and fractions.

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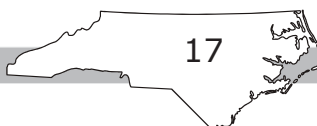
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- 26 Which statement **must** be true about a parallelogram?
- A It has 2 sets of parallel sides, and its opposite sides are the same length.
 - B It has 1 set of parallel sides, and its opposite sides are the same length.
 - C It has 2 sets of parallel sides, and all of its sides are the same length.
 - D It has 1 set of parallel sides, and all of its sides are the same length.
- 27 Which question will be answered with categorical data?
- A What are students' favorite foods?
 - B What is the number of pets a student has?
 - C How many brothers and sisters does a student have?
 - D How many games has a student's team won?
- 28 At a store, a package of 24 water bottles costs \$12. The store is advertising a sale for \$0.48 off the price of the package. What is the price of each bottle of water after \$0.48 is taken off the price of the package?
- A \$2.09
 - B \$2.08
 - C \$0.49
 - D \$0.48



Go to the next page.



29 Mr. Smith has two rectangular boxes.

- The first box is 20 inches long, 10 inches wide, and 12 inches tall.
- The second box has half the volume of the first box.
- The second box is 8 inches wide and 10 inches tall.

What is the length of the second box?

- A 13 inches
- B 14 inches
- C 15 inches
- D 16 inches

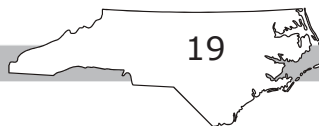
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30 What is the expanded form of 67.982?

- A $6 \times 10 + 7 \times 1 + 9 \times 0.10 + 8 \times 0.01 + 2 \times 0.001$
- B $6 \times 10 + 7 \times 1 + 9 \times 0.001 + 8 \times 0.01 + 2 \times 0.10$
- C $67 \times 10 + 9 \times 0.001 + 8 \times 0.01 + 2 \times 0.10$
- D $67 \times 10 + 9 \times 0.10 + 8 \times 0.01 + 2 \times 0.001$

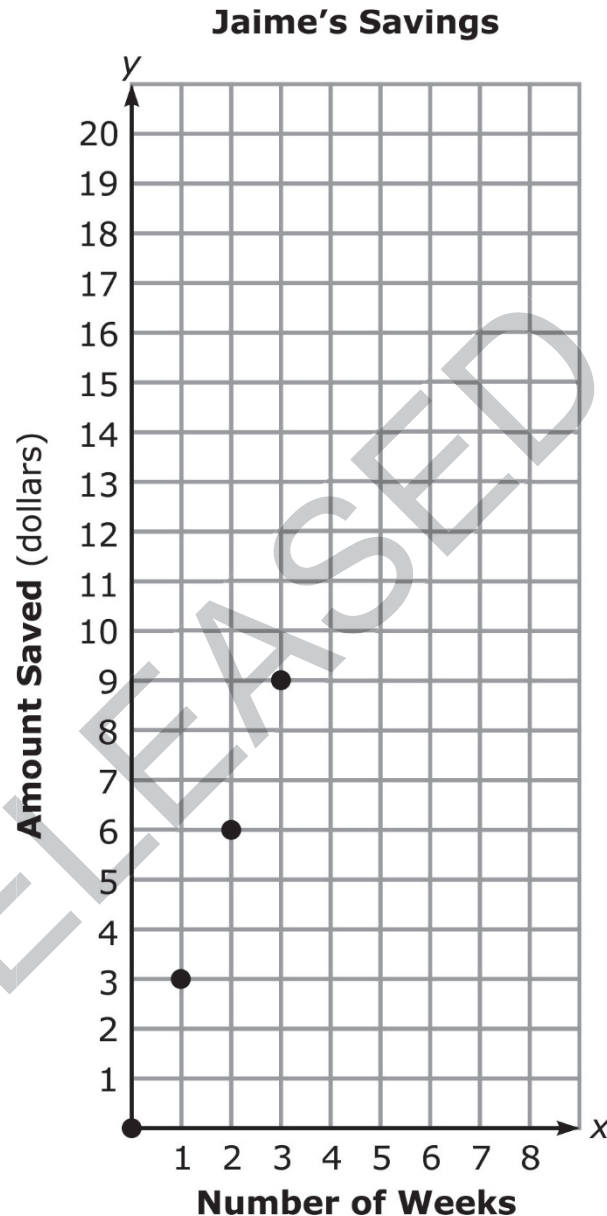
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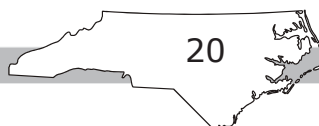
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- 31 Jaime saves the same amount of money each week. This graph shows how much money Jaime has saved in different weeks.



Question and answer choices A, B, C, and D are on the following page.



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How much money should Jaime have saved by week 6?

- A \$9
- B \$12
- C \$15
- D \$18

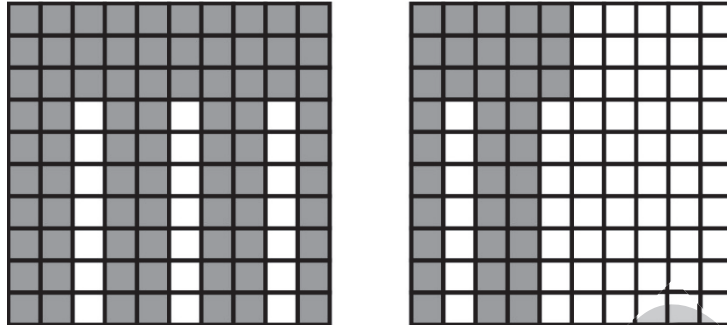
32 Eight students in an art club will share 50 pounds of clay equally. How many pounds of clay will each student get?

- A between 5 and 6 pounds
- B between 6 and 7 pounds
- C between 7 and 8 pounds
- D between 8 and 9 pounds

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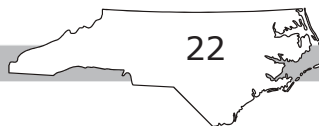


- 33 A figure is shown. The figure consists of two large squares. Each of the two large squares has an area equal to 1 square unit. Some parts of the figure are shaded.



Which expression represents the total amount of shaded area in the figure?

- A 0.23×5 square units
 B 23×5 square units
 C 0.023×0.005 square units
 D 0.23×0.5 square units
- 34 How many meters are in 27 kilometers?
 (Note: A kilometer is a thousand meters.)
- A 27
 B 270
 C 2,700
 D 27,000



Go to the next page.



35 What is the next pair of coordinates in this pattern?

(3, 4) (6, 8) (9, 12) (12, 16)

- A (13, 20)
- B (15, 16)
- C (15, 20)
- D (19, 20)

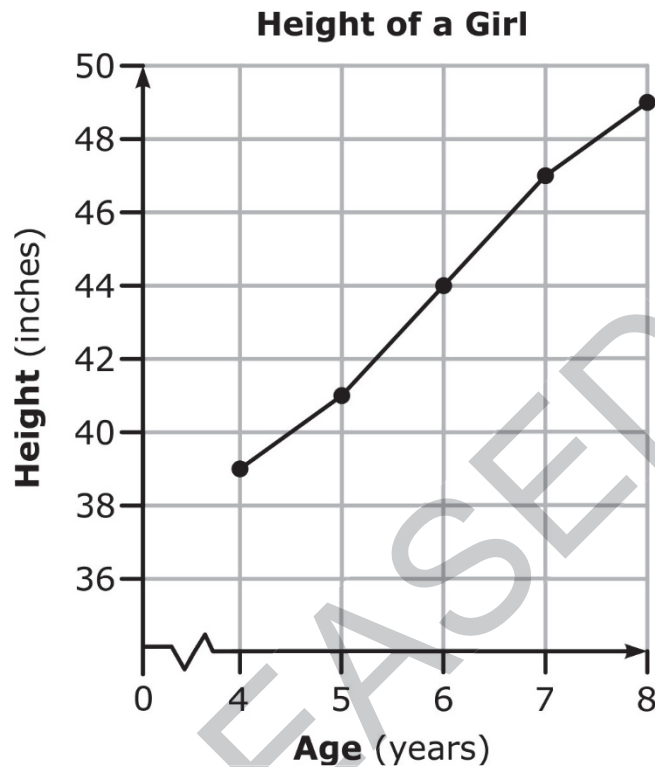
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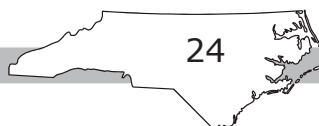


36 The height of a girl is recorded on this line graph.



How many inches did the girl grow between the ages of 5 and 8?

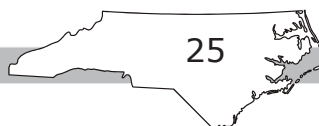
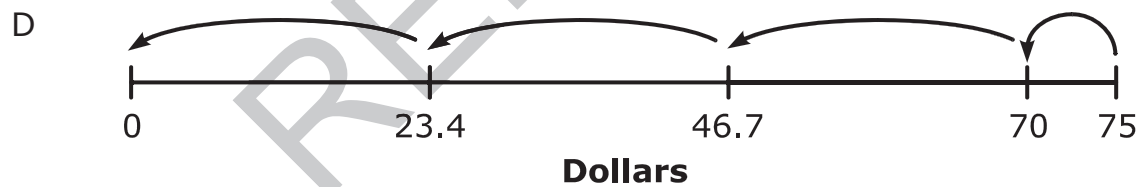
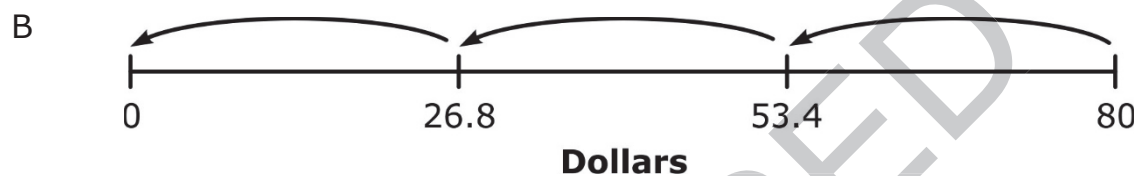
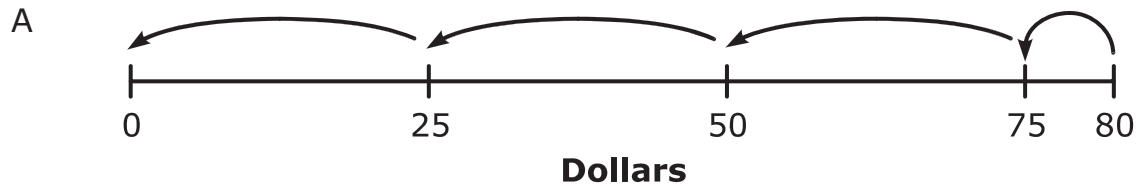
- A 7
- B 8
- C 9
- D 10



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- 37 Jake ordered 3 identical sleeping bags for a total of \$75, **not** including the cost of shipping. The shipping fee was \$5. Which number line shows how to calculate the cost of each sleeping bag and the cost of shipping?



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38 Annie tries to solve this problem.

$$16 \times (4 + 6) - 20 \div 4 = ?$$

She gets 35 as her answer. Is Annie's answer correct?

- A yes, because $140 \div 4 = 35$
- B yes, because a problem is solved from left to right
- C no, because she should have divided last
- D no, because she should have divided before she subtracted

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39 Nine students will equally share a 5-pound bag of marbles. How many pounds of marbles will each student get?

A $\frac{1}{45}$ pound

B $\frac{5}{14}$ pound

C $\frac{5}{9}$ pound

D $1\frac{4}{5}$ pounds

40 Luke has 4 feet and 7 inches of wire. Roger has 56 inches of wire. Who has more wire, and how much more?

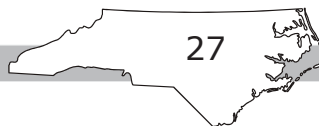
(Note: 1 foot = 12 inches)

A Luke, 7 inches more

B Roger, 1 inch more

C Luke, 1 inch more

D Roger, 7 inches more



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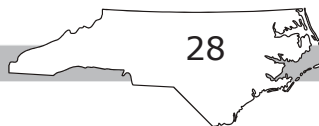


This is the end of the Mathematics test.

Directions:

- 1. Look back over your answers for the calculator active questions only.**
- 2. Make sure all your answers are entered appropriately in your book.**
- 3. Put all of your papers inside your test book and close your test book.**
- 4. Place your calculator on top of the test book.**
- 5. Stay quietly in your seat until your teacher tells you that testing is finished.**
- 6. Remember, teachers are not allowed to discuss questions from the test with you, and you are not allowed to discuss with others any of the test questions or information contained within the test.**

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GRADE 5 MATHEMATICS RELEASED FORM

Grade 5 Mathematics RELEASED Form 2026 Answer Key

Calculator Inactive



Item Number	Type	Key	DOK*	Standard
1	MC	D	2	5.NBT.5
2	MC	C	2	5.NF.1
3	MC	B	1	5.NF.4
4	MC	A	1	5.G.3
5	MC	C	2	5.NF.7
6	MC	D	3	5.OA.2
7	MC	B	2	5.NF.3
8	MC	D	2	5.OA.2
9	MC	B	1	5.NBT.6
10	MC	A	2	5.NF.1
11	MC	C	2	5.NF.4
12	MC	C	1	5.NF.1
13	MC	A	1	5.NBT.6
14	MC	D	1	5.NF.3
15	MC	B	1	5.NF.1
16	GR	216	1	5.MD.5
17	GR	78	2	5.NBT.6
18	GR	7/8	1	5.NF.1

GRADE 5 MATHEMATICS RELEASED FORM

19	GR	3/8	2	5.NF.4
20	GR	13870	2	5.NBT.5

Calculator Active



Item Number	Type	Key	DOK*	Standard
21	GR	16	2	5.NF.7
22	GR	1142	2	5.NBT.6
23	GR	36.18	2	5.NBT.7
24	GR	1/4	2	5.NF.4
25	GR	1/14	2	5.NF.7
26	MC	A	1	5.G.3
27	MC	A	1	5.MD.2
28	MC	D	2	5.NBT.7
29	MC	C	3	5.MD.5
30	MC	A	1	5.NBT.3
31	MC	D	2	5.OA.3
32	MC	B	1	5.NF.3
33	MC	A	1	5.NBT.7
34	MC	D	1	5.MD.1
35	MC	C	2	5.OA.3
36	MC	B	2	5.MD.2
37	MC	A	2	5.NBT.7
38	MC	D	3	5.OA.2
39	MC	C	2	5.NF.3
40	MC	B	1	5.MD.1

GRADE 5 MATHEMATICS RELEASED FORM

***DOK:**

1 = Recall

2 = Skill/Concept

3 = Strategic Thinking

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